

Module 5 : Lightning Flow and approval Process

5.1 Approval Process

Definition & Use Case

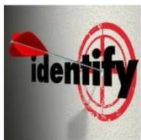
- Approval processes are multi step or single step actions performed which require user confirmation to process a record.
- Let's understand why approval process is needed in Salesforce.



- In our Recruiting Application, any New positions created as per company's requirement must be approved internally by the Recruiting Team before its released for the recruiting process.

5.1 Approval Process

Sequence of Steps in Approval Processes



Process Definition

Global Characteristics
Validates
Which records will undergo the process
Restrictions that is applied on records



Initial Submission

Workflow Actions
Performs action when Process is started



Step Definition

Decision Criteria & Approval assignment
Depending on criteria what actions should the record undergo



Final Approval

Workflow Actions
Actions to be taken when record is approved



Final Rejection

Workflow Actions
Actions taken when the record in process is rejected



Recall Action

Workflow Actions
Actions to be taken when validation on record is recalled

5.1 Approval Process

Wizards to create Approval Process

- **Standard Setup Wizard**

- This wizard is more comprehensive and allows more detailed configuration of approval process.
- We can add multiple steps with options to customize the criteria at each step and various actions to be taken at each step
- This wizard provides the granular control of approval steps set up deciding who and how many approvers must be added and also order of the approvals.
- Suitable for complex business requirements , where custom actions needs to be defined at each step
- We can define recall actions, final approval and rejections

5.1 : Multi Step approval process

Requirement 1.1

All the 'New' Positions must undergo an approval process from Recruiter (Mario Ruiz).

- Use Jump Start Wizard
- Make the below fields available for the approvers to be visible

Position Name	Job Description	Location
Department	Date Opened	Pay Grade
Salary Range	Status	Sub-status

- Send email to Approver Mario Ruiz if Approval request is initiated on a record

5.2

Introduction to Lightning Flow

Lightening Flow: It is an **automation** Tool. Powerful tool

Earlier we have 3 automation tools

1. Workflow rules – not used now
2. Process builder - not used now
3. Flow

2 types of flow:

1. **Screen Flow:**
2. **Record Triggered Flow:**

Automatic Actions : Sending an email automatically.

DML : insert, update, delete

Based on these DML statements, automatic action like email can be triggered.

SQL Triggers - without coding

In salesforce also, we have **Apex triggers** : we use it to implement complex functionality (which cant be done with Flows), can implement it with Apex triggers.

5.2 Introduction to Lightning Flow

Salesforce Flow Definition

- A **Flow** is business process collects data and uses it to process certain action which is automated.
- **Flow Builder** is a graphical tool for creating & managing flows.
- **Salesforce Lightning Flow** is a process automation feature which allows us to create, manage and run automatic actions/tasks by clicks.
- Flows can be initiated by users by clicking on button or triggered automatically by certain events or actions.
- Easily be integrated with other tools like approval process, Apex code & external systems.
- Flow can have maximum 50 versions with one Active version.
- The main advantage of Salesforce flow includes increased efficiency by reducing the manual tasks, accuracy in data handling and boost customer experience by speeding the processes.

Salesforce Flow & types

Salesforce Flow includes several types of automated processes, few frequently used are:

- **Screen Flows**
 - Guide users through a series of steps via custom screens
- **Record Triggered Flows**
 - It starts when a record is created, updated or deleted and performs various actions in backend
- **Auto Launched Flows**
 - Runs in background without any user interaction
- **Scheduled Trigger Flows**
 - These are triggered at specific time

5.3 Screen Flow

Definition & Use Case of Screen Flows

Screen Flows are designed to guide users through various of steps like filling out forms, making decisions, step-by-step processes etc.,

Purpose of screen flows:

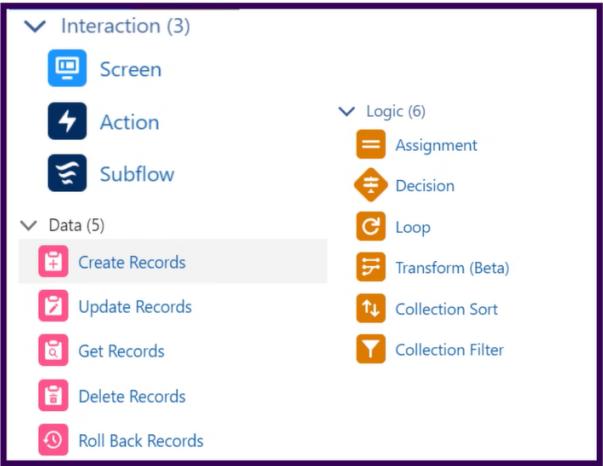
- Consider external Users who do not have access to Recruiting Application or the Objects, & would like to add their profile details in Candidate Object to apply for Jobs suitable for their Positions.
- We create a platform for these Users to enter their basic details, address information, if experienced then Current Employer details will be entered by these Users.
- We collect all the information from these Candidates/external Users via different screens.
- The Navigation to different screens are can be controlled in the Screen Flow.
- All the collected information about the Canidates are then finally added to the Candidate Object.

Introduction to Elements , Variables & Resources

Elements

- These are building blocks of flow used to create logic and functionality
- These are categorized into three:

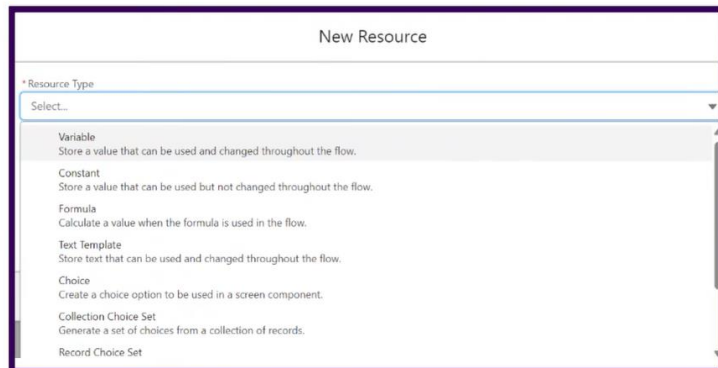
Interaction	Logic	Data
Screen	Assignment	Create
Action	Decision	Update
Sub-Flow	Loop	Delete
	Collection	Rollback
		Get



Introduction to Elements , Variables & Resources

Resources

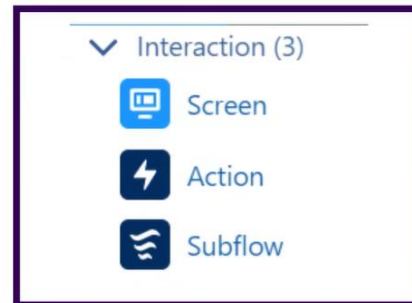
- These are containers to store and manipulate data in the flow.
- Types
 - Variable
 - Constant
 - Formula
 - Text Template
 - Choice
 - Choice Set
 - Collection
 - Record
 - Picklist
 - Stage



Elements

• Interaction

- Screen: When user inputs or interactions are required to run the process
- Action: When certain actions needs to be started outside the flows
- Sub-Flow: When we want another flow to run creating dependency to complete the process



Variables

Variables

- Used to hold values that can be manipulated in flow
- Types of Variables based on data types:
 - Text
 - Record
 - Number
 - Currency
 - Boolean
 - Date
 - Boolean
 - Date/Time
 - Data/Time
 - Picklist
 - Multi-Picklist
 - Apex Defined

New Resource

Text
Record
Number
Currency
Boolean
Date
Date/Time
Picklist

Select...

☐ Allow multiple values (collection) ⓘ

A value is required.

Demo

5.3 Create Screen Flow Using Resources

Requirement 1.1:

Create a Screen Flow to take inputs on Candidate Details for below given data fields:

Element	Screen 1	Screen 2
Section	Candidate Details	Address Information
Fields/Components	First Name Last Name Phone Email Gender	Address (Component)

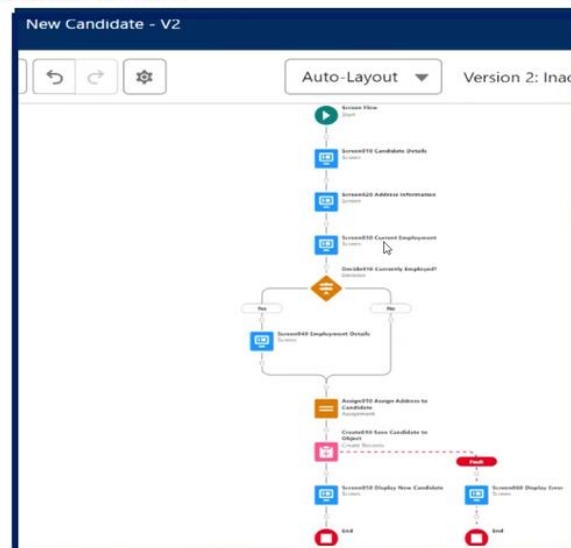
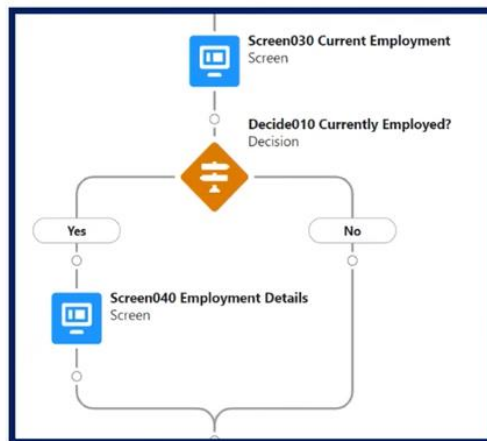
NOTE: Remove the Previous & Pause buttons on Screen 1.

In Lightning Experience, we have 3 types of pages:

1. Home Page
2. App Page
3. Record Page

Demo

5.3 New Version of New Candidate Details using Decision Element



Record Triggered flow:

1. Object on which the flow will be created.
2. When the flow should be triggered (record is created, record is updated, record is deleted, record is created or updated)
3. Entry conditions
4. Actions(Automatic actions)

4. Actions(Automatic actions) --> send email, update records, create a record, call a flow, call approval process, post to chatter,

5.4 Record Triggered Flow

Define and Use Case

- Record-triggered flows are automated processes in a system that kick off (or are triggered) when specific changes or events occur within a record or object.
- They automate repetitive tasks, saving time and effort keeping the application efficient
- Respond immediately to change in data
- User have flexibility to design custom logic & actions to meet their business requirements
- Data validation and manipulation is achieved using Record Triggered flows
- They can be easily integrated to any kind of platform or environment
- Use Case:
 - When a new Candidate applies for a position/job application in a company
 - Based on the candidate skills, trigger an interview process automatically notifying both candidate & interviewer

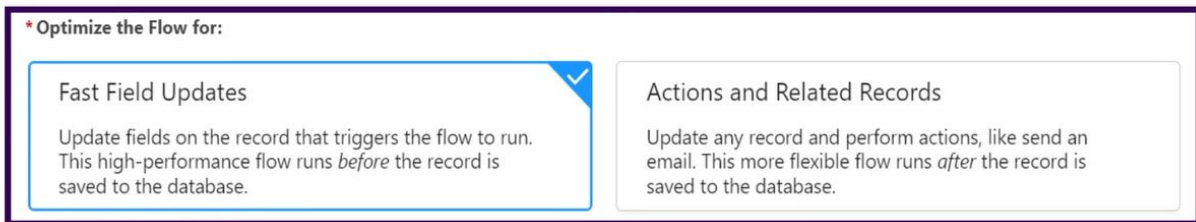
Types of DML

- DML abbreviated for “Data Manipulation Language” is used to make our computer understand to load , change or unload data from Salesforce.
- Standard DML statements used for data changes are:

DML Statements	Use Case
INSERT	Used add new data to database
UPDATE	Used to make changes on existing data
UPSERT	Used when we want to add new data along with changes in existing data
DELETE	Used to remove data from database
UNDELETE	Used to retrieve data from recycle bin to Salesforce database
MERGE	Used to remove duplicates and combine data

Introduction to Before Trigger Record Triggered Flow & After Trigger Record Triggered Flow

- DML statements are initiators of Record Trigger flows
- A flow can be initiated with change in data/values (a cause of DML action) in database
- Triggers lead to perform certain tasks/actions in Salesforce platform. This can happen in 2 ways.
 - Before Trigger Flow: Used to perform actions before a data is changed via DML
 - After Trigger Flow: Used to perform actions after a data is changed via DML



Types of Record Triggered flow:

1. Before Save Record triggered flow

--> this flow will be executed before the record is saved to the database.
To create this flow, select 'Fast Field Updates' option.

2. After Save Record Triggered flow

--> this flow will be executed after the record is saved to the database. 
To create this flow, select 'Actions and Related records'.

Before save triggered flow:

Use cases:

1. FIELD Validations (just like validation rules)
2. To update any field

Introduction to Before Trigger Record Triggered Flow & After Trigger Record Triggered Flow

- Triggers can be imposed on the desired object's data when there is a change in data via DML action:
 - When new data is added to database/selected Object via INSERT
 - When existing data is changed via UPDATE
 - When UPSERT action is performed on the database/Object desired
 - When DELETE action is performed on data from Object selected

Select Object

Select the object whose records trigger the flow when they're created, updated, or deleted.

* Object

Configure Trigger

* Trigger the Flow When:

☒ A record is created

☐ A record is updated

☐ A record is created or updated

☐ A record is deleted

5.4 Record Triggered Flow

Before Triggered Flow

- When the requirement states to execute actions before when a record is saved, we optimize the flow with "Fast Field Updates"
- This supports all the trigger options such as INSERT, UPDATE, UPSERT, DELETE.
- The DELETE is supported only in this trigger flow

Toolbox

Elements Manager

Interaction (1)

Custom Error

Logic (5)

Assignment

Decision

Loop

Collection Sort

Collection Filter

Data (2)

Update Records

Get Records

* Optimize the Flow for:

Before the record is deleted

The flow can access any record and perform actions before the triggering record is deleted from the database. To create, update, or delete other records, add Create Records, Update Records, and Delete Records elements to your flow. The \$Record global variable contains the Salesforce record that launches the flow.

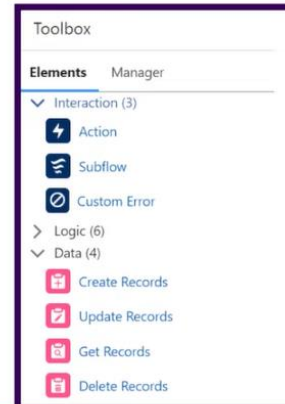


If a deleted record is later recovered, the flow's database changes and actions aren't rolled back. Also, record recovery can't trigger a flow to run. If you require automation for when records are recovered from the Recycle Bin, consider using an Apex after undelete trigger.

5.4 Record Triggered Flow

After Triggered Flow

- When the requirement states to execute actions after when a record is saved, we optimize the flow with “Actions & Related Records”
- This supports the trigger options such as INSERT, UPDATE, UPSERT.
- DELETE trigger is NOT supported in this flow because the actions will be related to the triggering records.



***Optimize the Flow for:**

Fast Field Updates

Update fields on the record that triggers the flow to run. This high-performance flow runs *before* the record is saved to the database.

Actions and Related Records

Update any record and perform actions, like send an email. This more flexible flow runs *after* the record is saved to the database.

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Demo

5.4 After Save Record Triggered Flow on Position Object Requirement 1:

Create a Record Triggered Flow to Send Email to Recruiting Manager (Phil Katz) for all Closed Positions

Step 1: Create Email template

Email Template Detail		Edit	Delete	Clone
Email Templates from Salesforce	Unfiled Public Classic Email Templates			
Email Template Name	Position Closed			
Template Unique Name	Position_Closed			
Encoding	Unicode (UTF-8)			
Author	Srividya.M [Change]			
Description				
Created By	Srividya.M, 13/05/2024, 12:27 pm			
		Edit	Delete	Clone

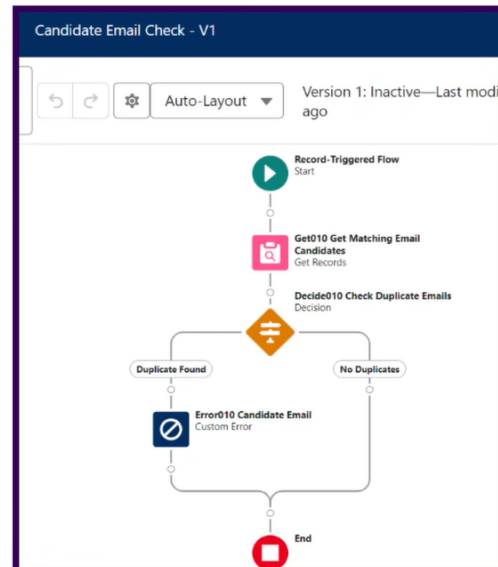
Email Template		Send Test and Verify Merge Fields
Subject	Position is Closed...	
Plain Text Preview	Hi {!User.Name}. The Position {!Position__c Name} is CLOSED. Thanks, Recruitment Team	

Demo

5.4 Before Save Record Triggered Flow

Requirement 3:

- Create a Record triggered Flow to check if Candidate's Email is filled in.
- Retrieve records & check for duplicate emails
- If Duplicates are present, then throw error stating "The Entered Candidate Details with {!\$Record.Email__c} already Exist. Kindly check the details."



Types of Record Triggered flow:

1. Before Save Record triggered flow

--> this flow will be executed before the record is saved to the database.
To create this flow, select 'Fast Field Updates' option.

Use Cases:

1. Field Validations(just like validation rules)
2. to update any field

2. After Save Record Triggered flow

--> this flow will be executed after the record is saved to the database.
To create this flow, select 'Actions and Related records'.

Use Cases: send email, create a record, call a flow, call approval process, post to chatter